W-R-101F
March 13, 1970
SUPERSEDING
Fed Spec W-R-101E
December 10, 1964

FEDERAL SPECIFICATION

RANGES, ELECTRIC, DOMESTIC

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers electric free-standing ranges and slide-in-ranges used as cooking appliances.

1.2 Classification.

1.2.1 Types, styles, and sizes. The ranges shall be of the following styles and sizes 1/, as specified (see 6.2).

Type I - Ranges, free standing 1/

Style 1 - Single oven

Size 2 - 36 inches wide
Size 3 - 30 inches wide
Size 4 - 24 inches wide
Size 5 - 20 inches wide

Style 2 - Double oven 40 inches wide

Type II - Ranges, built-in (slide-in) 1/

Size 2 - For a nominal 30 inch-wide opening 2/
Size 3 - For a nominal 27 inch-wide opening 2/
Size 4 - For a nominal 24 inch-wide opening 2/
Size 5 - For a nominal 20 inch-wide opening 2/

1/ A tolerance of plus or minus 1/4 inch wide is permissible.
2/ Opening width defines product width.

FSC 7310

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:

MIL-R-11337 - Ranges, Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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Underwriters' Laboratories, Inc. (UL) Standard:

No. 858 - Household Electric Ranges.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc. 207 East Ohio Street, Chicago, Illinois 60611; 1285 Walt Whitman Road, Melville, L. I., New York 11746; or 1655 Scott Blvd., Santa Clara, California 95050.)

American Society for Testing and Materials (ASTM) Standards:

B 407 - Nickel-Iron-Chromium Alloy Seamless Pipe and Tube.
C 282 - Standard Method of Tests for Acid-Resistance of Porcelain Enamel.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

American National Standards Institute (ANSI) Standard:

C 71.1 - Household Electric Ranges.

(Application for copies should be addressed to the ANSI Standards Institute, Inc., 1430 Broadway, New York, New York 10018.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), before production is commenced, a preproduction sample range being procured under this specification shall be submitted or made available to the contracting officer or his authorized representative for approval, in accordance with 4.2. Approval of the preproduction sample authorizes commencement of production but does not relieve the supplier of responsibility for compliance with the detailed provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items. Whenever a material, design or construction change or deviation is made to the ranges after approval of the preproduction sample, the Government shall withhold acceptance of subsequent ranges incorporating those changes until a sample containing changes or deviations has been inspected and approved in accordance with 4.2 or the contracting officer has given written approval based on the evidence furnished that the changes or deviations made to not adversely affect the design and performance of the ranges. The preproduction unit shall be kept available during entire production and may be furnished as the last unit of the contract. However, before shipment the preproduction unit shall be reworked to meet all requirements of this specification.

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3.2 Standard product. The ranges delivered under this specification shall be the manufacturer's standard commercial product, except for any changes necessary to comply with specification requirements. All ranges of one style and size furnished under this specification for any one contract, including parts and assemblies thereof, shall be interchangeable.

3.3 Standards. The ranges shall conform to the applicable requirements of UL Standard No. 858 and ANSI Standard No. C 71.1 - 1964.

3.3.1 UL certification. Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the range he proposes to supply under this specification meets the applicable requirements of UL Standard No. 858. Acceptable evidence of meeting the applicable requirements of this standard shall be the UL Label, UL Listing Mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the ranges have been tested and conform to UL Standard No. 858.

3.3.2 ANSI certification. Prior to approval of the preproduction sample, if one is submitted, or prior to approval of the first shipment, the supplier shall submit satisfactory evidence to the contracting officer or his authorized representative that the model range he proposes to furnish under this specification meets the requirements of ANSI Std. C 71.1-1964. Acceptable evidence of meeting the requirements shall be the manufacturer's certified data or a certified test report from a recognized independent testing laboratory acceptable to the Government indicating that the model range offered has been tested and conforms to ANSI Std. C 71.1-1964.

3.4 General requirements. The ranges shall be of the types, styles, and sizes stated in 1.2.1.

3.4.1 Electrical characteristics. Unless otherwise specified (see 6.2), the ranges shall be designed for operation on a nominal 120/240-volt, 60-Hertz, 3-wire, single phase alternating current system.

3.4.1.1 Connections and accessibility. All terminal connections and wiring to switches shall be such that repairs can be made without disassembly of switch structures or other electrical components. Screwed terminals and push-on connectors shall be allowed on solid and stranded wire, 12 gage and smaller. Push-on connectors used on solid wire, 14 gage and larger may be soldered to ensure good electrical contact. Insulation piercing terminals shall be allowed on stranded wire, 16 gage and smaller. There shall be no loose strands over 3/16 inches long. The maximum allowable number of loose strands under 3/16 inches long shall be three.

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3.4.1.2 Appliance outlets. Appliance outlet receptacles shall receive standard grounding type plug caps with 3-prong parallel blades located at convenient points on the ranges.

3.4.2 Ovens. Oven linings shall be assembled by welding, bolting, riveting, or metal self-locking fasteners. Oven insulating material shall be of the blanket type installed in accordance with the manufacturer's standard practice and shall prevent surface temperature from exceeding those specified in 3.4.4. Loose fill insulation shall not be used. The heating element for the oven heating units shall be incased in a seamless metal tube as specified in 3.4.9.2. The bake wattages and broil wattages shall be as specified in 3.5.1.2. Oven heating units shall be of the bayonet or blade type or be capable of being replaced by the removal of not more than four screws, exclusive of the terminal screws and screws for fastening the wire trough or back plate. The connection to the oven heating system shall be located outside the oven and shall be protected from grease or moisture from the oven. The heating units shall be installed to permit cleaning of the oven. It shall not be necessary to interchange the position of an oven heating unit for either baking or broiling. Baking ovens shall have at least two adjustable racks with rack stops and the racks shall be made of corrosion-resisting steel, or nickel-plated or chrome-nickel-plated steel. An oven intended for broiling shall be equipped with a smokeless type broiler pan and rack of corrosion-resistant material or finish. The rack shall not jam when placed into, or removed from the shelf supports.

3.4.2.1 Oven doors. All oven doors of single-oven ranges and the main oven of the double-oven range shall be a drop-opening type. Springs for counterbalancing shall be located entirely outside the heated zone or protected from heat so usability is not affected. All doors shall be equipped with a handle and means for holding the door closed. The surface temperatures of ranges shall not exceed the requirements of 3.4.4. Drop doors shall be balanced to remain fully open without securing when in the fully open position permitting full access to the usable oven area. Ranges equipped with removable oven doors will be acceptable. When specified (see 6.2), a heat resisting viewing window oven door shall be provided.

3.4.2.2 Oven light. Unless otherwise specified (see 6.2), all ovens, except ovens for size 5 ranges shall be provided with a light to illuminate the oven interior. The light shall be located or protected to prevent being damaged. The oven light may be actuated by opening the door or by use of a separate switch except a separate oven light switch shall be provided when a heat resisting viewing window oven door is furnished.

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3.4.2.3 Temperature control. The temperature of the oven compartment shall be controlled by a snap-action thermostat and a separate switch or by a thermostat with an "OFF" position. The "OFF" position shall be marked on the thermostat dial. The thermostat actuating components shall be protected against moisture, grease, and direct heat radiation. A detent or other means shall be provided on all controls to prevent the control from being accidentally turned on.

3.4.2.4 Oven time control. Unless otherwise specified (see 6.2), all ovens except ovens for size 5 ranges shall be provided with a 60-minute or more oven time control. The oven time control shall be equipped with an alarm signal which will sound whenever the preset time has been completed (see 4.4.1). The time control range shall have the graduations in minute intervals.

3.4.3 Surface heating units. The ranges shall be provided with 8 inch and 6 inch nominal diameter surface heating units. The 8 inch nominal diameter surface heating units shall have a minimum rating of 2,000 watts and 6 inch nominal diameter surface heating units shall have a minimum rating of 1,200 watts each. The heating element for the surface units shall be incased in a seamless metal tube as specified in 3.4.9.2. The surface heating units shall be hinged or easily removable and easily replaceable to permit cleaning. Means shall be provided to allow replacement of the hinged surface heating units by use of hand tools. A reflector pan shall be furnished under each surface heating unit. Range tops shall allow proper seating of the surface units and spillage collectors.

3.4.3.1 Surface unit switch. Each surface heating unit shall be individually controlled by a rotary or multiple-push-button switch except controls shall not be mounted on cooking tops as defined in ANSI C 71.1. The switches shall provide at least six-position, five-heat control and one "OFF" position. The various positions shall be legibly and permanently marked on or immediately adjacent to each control. A detent or other means shall be provided on all controls to prevent the control from being accidentally turned on.

3.4.4 Surface temperatures. When tested as specified in 4.4.2, the surface temperature rise on the exterior surfaces of the ranges shall not exceed the values given in table I.

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TABLE I. Surface temperatures

Component part	Maximum rise above room temperature (degrees Fahrenheit)
Door panel	
Glazed 1/	110
Metal	95
Exterior sides and top	110
Door handle	40
Knobs	40

1/ No temperature shall be taken on the metal closer than 1 inch to the opening framing the glazed panel.

3.4.5 Manuals.

3.4.5.1 Civil agencies. The supplier shall furnish his standard manual (see 6.2) containing installation, operating, and maintenance instructions and a list of spare parts in duplicate with each range.

3.4.5.2 Instruction manual requirements (GSA procurement for military use). Unless otherwise specified (see 6.2), the supplier shall furnish with each end item (1) copy of a instruction manual which, as a minimum, shall contain complete installation, operation, maintenance and repair instructions, including applicable charts and diagrams and an illustrated parts data section. Operation and maintenance instructions shall also include preventative maintenance service instructions as well as a trouble shooting chart or guide. The instructions portion of the manual shall be in simple language, easily understood by personnel responsible for the installation, operation, maintenance and repair of the end item. Subjects difficult to narrate shall be clarified by the use of illustrations. The parts data section of the manual shall include, as a minimum, a list of all end item components, assemblies and those parts which require replacement for maintenance of the end item. It need not include items which cannot be disassembled or are of such design that disassembly is impractical. Each of these components, assemblies, and parts shall be illustrated, except that items of common hardware and parts having a normal life expectancy equal to that of the end item may, but need not be listed or illustrated. The parts list shall indicate, as a minimum (1) supplier's part numbers, (2) description of parts, and (3) for parts not manufactured by the supplier, the actual manufacturer's name, address, and part number. Each listed component, assembly, and part shall

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be individually keyed to illustrations clearly depicting its application in the end item. Illustrations shall be photographs or line drawings, sharp in detail to permit ready identification of parts. The part number assigned shall be definitive and self supporting so that specific part can be ordered by citing only part number and noun description. The supplier shall overpack with the manual supplemental data, when furnished by the Government. The instruction manual shall be bound in a single protective cover.

3.4.5.2.1 Instruction manual approval. Unless otherwise specified (see 6.2), the supplier shall submit 4 copies of the proposed instruction manual to cognizant contracting officer for review by the using military service to determine compliance with 3.4.5.2. Time frames for manual submission, review, and notification or approval or revisions required shall be as negotiated by the contracting officer.

3.4.5.2.2 Model and serial numbers. Unless otherwise specified (see 6.2), the supplier shall assign a specific model number to all identical end items furnished under any one contract and, in addition, serial numbers will be assigned under each respective model produced. Model and serial numbers, with other data required, shall be affixed to each item. The supplier shall stencil or print the model and serial number on the exterior of each container in accordance with MIL-STD-129. This information is required in addition to other provisions of MIL-STD-129 and will be identical to the model and serial numbers affixed to the end item. On or before final shipment, the supplier will forward a list of model and serial numbers of each item furnished under the contract to the contracting officer for distribution to the using military service.

3.4.6 Indicating light. Unless otherwise specified (see 6.2), each range shall be equipped with a light to indicate when the surface unit(s) or oven(s) are energized.

3.4.7 Leveling devices. All ranges shall be equipped with at least two leveling devices capable of leveling the ranges in accordance with the manufacturer's standard practice.

3.4.8 Marking for identification. The ovens shall be permanently marked with the serial and model number and with the manufacturer's name, trade name or trademark of such known characters as to be readily identifiable by the manufacturer. In addition, electrical characteristics and such information as required by UL Standard No. 858 shall also appear on the nameplate.

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3.4.9 Materials. Materials not definitely specified shall be of the quality normally used by the manufacturer for his range provided the completed item complies with all provisions of this specification. Range cooktops, door panels, and side and other front panels shall be fabricated of material suitable for the finish specified in 3.4.10. Spillage collectors shall be fabricated from steel suitable for porcelain enameling, corrosion-resisting steel, aluminized steel, aluminum, nickel plated or chrome nickel-plated steel. Oven linings shall be fabricated from steel suitable for porcelain enameling, corrosion-resisting steel, or chrome or nickel-plated steel.

3.4.9.1 Plastic. Plastic for knobs, door handles, and drawer handles shall be molded, heat-resistant, thermosetting type and shall show no breakage, chipping, or cracking visible to the naked eye when subjected to the drop test specified in 4.4.3.

3.4.9.2 Seamless metal tube. The seamless metal tube shall conform to ASTM B-407-65T.

3.4.10 Finish. The ranges shall be white in color. Cooktops and oven door panels shall have a porcelain enameled finish. Side and other front panels shall be finished in porcelain enamel or a suitable high temperature organic finish. Side and front panels provided with an organic finish shall be given a rust-resistant primer coating on inside surfaces. Porcelain enamel shall be glossy and free from discolorations, cracks, thin spots, holes, burn-off, chipping, copper head, fish scaling, or enamel lumps i.e. wrinkled or wavy or other imperfections, and shall not show bare metal at corners. Ground coat shall not show through cover coat on exterior surfaces. Porcelain enamel shall not have any evidence of crazing, crawling, specking or tearing which will affect appearance. Porcelain enamel on cooktops shall have a class A or better acid resistance when tested in accordance with ASTM C 282-67 (see 4.3.2.2). Porcelain enamel thickness shall be not greater than 12 mils when tested as specified in 4.4.4. When porcelain enameled oven linings are provided, the enamel shall be applied to interior surfaces of the oven linings including welded joints. Surfaces of parts not exposed to view in normal usage shall be protected in such a manner that any oxidation which may occur will not interfere with its intended function. An oxide coating is acceptable for these unexposed surfaces. Exposed corrosion-resisting steel shall be finished in accordance with the manufacturer's standard practice.

3.4.11 Operation. Controls, switches, and moving parts of the ranges, shall not bind nor be loose, and electrical circuits shall operate through a complete cycle without malfunction when tested as specified in 4.4.1.

3.5 Detailed design and construction. The ranges shall conform to the Installation Form 0 classification of UL Standard No. 858.

3.5.1 Type I ranges, free standing.

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3.5.1.1 Surface units. Type I ranges shall be equipped with surface units and switches as specified in 3.4.3 and 3.4.3.1, respectively. The size 2 and 3 ranges shall be equipped with two 8-inch and two 6-inch surface units or one 8-inch and three 6-inch surface units. The sizes 4 and 5 ranges shall be supplied with four surface units as stated above or when specified (see 6.2) with one 8-inch and two 6-inch units. When specified (see 6.2), one of the units on the style 1, size 2 or 3 and the style 2 range shall be thermostatically controlled.

3.5.1.2 Ovens. Style 1 ranges shall be equipped with one oven and style 2 range shall be equipped with two ovens. Ovens shall conform to the requirements of 3.4.2 through 3.4.2.4 inclusive. Broil wattage shall be rated minimum 3000 watts for style 1, size 2, 3, and 4 ranges and one oven of the style 2 range. Broil wattage of the style 1, size 5 oven shall be rated minimum 2500 watts. Bake wattage shall be minimum 2400 watts for all ovens except style 1, size 5, and the second oven of the style 2 range both of which shall be minimum 2000 watts. Each oven shall be equipped with two heating units, one located at the top and one at the bottom of the oven space. The oven on the style 1 ranges shall have not less than 10 inches usable oven space between the heating units. The oven shall have a minimum volume of 5,000 cubic inches for size 2, 4,500 for size 3, 4,000 for size 4, and 3,500 for size 5 including the space occupied by the heating units. One of the ovens on the style 2 ranges shall have not less than 10 inches of useable oven space between heating units and a volume of not less than 5,000 cubic inches. The second oven on the style 2 range shall have a volume, including the space occupied by the heating units, of not less than 2,500 cubic inches and shall have not less than 10 inches useable space between the heating units. If the space is divided into two distinctly and permanently separated compartments for baking and broiling, each space shall have not less than 5 inches useable space between the heating unit and top or bottom of the compartment.

3.5.1.3 Oven clock. The style 2 range shall be equipped with an electric oven timing clock which will turn the 5,000 cubic inch oven on and off at set time. The range shall also be provided with an interval timer for accurate timing from 0 to 60 minutes or more with the graduation in minute intervals. An alarm signal shall sound when selected time has elapsed (see 4.4.1). A combination oven timing clock and interval timer is acceptable.

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3.5.1.4 Appliance receptacles. Style 1, size 2 and 3, and style 2 ranges shall be equipped with at least one appliance receptacle as specified in 3.4.1.2. When specified (see 6.2), additional appliance receptacles shall be provided and at least one of them shall be time-controlled if the oven is time-controlled. When specified (see 6.2), the style 1, sizes 4 and 5 ranges shall be equipped with one appliance receptacle.

3.5.2 Type II ranges, built-in (slide-in).

3.5.2.1 Type II surface units. The type II ranges shall be equipped with surface units and switches as specified in 3.4.3 and 3.4.3.1, respectively. The size 2 and 3 ranges shall be equipped with two 8-inch and two 6-inch surface units or one 8-inch and three 6-inch surface units. The sizes 4 and 5 ranges shall be equipped with four surface units as stated above or when specified (see 6.2) with one 8-inch and two 6-inch units. When specified (see 6.2), one of the units on the sizes 2 and 3 ranges shall be thermostatically controlled.

3.5.2.2 Type II oven. The sizes 2 and 3 ranges shall be equipped with one oven having not less than 10 inches useable space between the top and bottom heating units and, including the space occupied by the heating units, shall have a volume not less than 5,000 cubic inches including the space occupied by the heating unit. The heating units shall be located as to enable the oven to be used for baking and broiling. Sizes 4 and 5 ranges shall be equipped with an oven as specified above except that the volume shall be not less than 4,000 cubic inches. The bake wattage shall conform to the requirements of ANSI Std C71.1 and the broil wattage shall be rated at a minimum of 2,500 watts.

3.5.2.3 Type II oven door(s). The oven door(s) for the type II range shall be as specified in 3.4.2.1. The oven door shall be recessed and the sides of the range cabinet shall project past the oven door.

3.5.2.4 Optional equipment. Each size of type II range shall have appliance receptacles in the options as specified for each comparable size of type I ranges in 3.5.1.4.

3.6 Workmanship. The finished range shall be clean and shall not contain any burrs, slivers, sharp edges or rough die, tool, gouge or grind marks. Holes for bolts, screws, rivets, or threaded fasteners shall be accurately punched or drilled and any obstruction that prevents proper seating of rivets shall be removed, the driven rivets shall be fully peened with the heads in full contact with riveted members. Components shall not be fractured, buckled, kinked, bent, dented, punctured, or malformed. There shall be no evidence of pits or stains, blistering, flaking, peeling or large discoloration (exposed surfaces). Components shall be free of area(s) of no film or of rust, foreign matter imbedded in finish or finish wet or tacky (painted surfaces), sags or runs.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required it shall be examined for defects listed in table II, dimensions specified, and tested in accordance with 4.4.1 and 4.4.2. Failure to pass any of the tests shall be cause for rejection of the preproduction sample. The presence of any defects listed in table II or any dimension not within specified requirements shall be cause for rejection of the preproduction sample

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended modified, or qualified in this specification or applicable purchase document.

4.3.2 In-process inspection.

4.3.2.1 Process examination. Examination shall be made of the following operations to establish conformance to specified requirements. Whenever nonconformance is noted, correction shall be made to affected items and lot in process.

(a) Finishes. Finish operations on unexposed surfaces shall be examined for compliance with 3.4.10.

(b) Thermal insulation. Installation of insulation shall be examined for compliance with 3.4.2. Particular attention shall be given to type of insulation and method of application.

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4.3.2.2 Intermediate testing of cooking tops. Porcelain enameled finished cooking tops shall be tested to determine compliance with the acid resistance requirements of 3.4.10. Porcelain enamel cooking tops, oven door panels and side panels shall be tested to determine compliance with the thickness requirements specified in 3.4.10. The lot shall consist of one day's production of cooking tops, oven door panels, front panels and side panels, as applicable. The sample unit shall be one cooking top, oven door panel, front panel and side panel, as applicable. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0 defects, expressed in terms of defects per hundred units. Test procedure for acid resistance shall be in accordance with ASTM Specification C 282-67. Test procedure for thickness shall be as specified in 4.4.4.

4.3.2.3 Intermediate testing of knobs and handles. Knobs and handles shall be tested as specified in 4.4.3. The lot shall consist of all knobs and handles, as applicable, offered for inspection at one time. The sample unit shall be one knob or handle. The inspection level shall be S-1 with an AQL of 6.5 defects, expressed in terms of defects per hundred units.

4.3.3 End item inspection. The inspection lot shall consist of all ranges of one type, style, and size offered for inspection at one time. The sample unit shall be one completely fabricated range.

4.3.3.1 Visual examination. Examination shall be made of the range for the defects in table II. The inspection level shall be level II with an AQL of 2.5 for major defects and 6.5 for total defects expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish (general)	Not as specified.		X
	Evidence of pits, or stains, peeling, flaking, blistered, spots, large discolorations (exposed surfaces).	X	
	Not finished, plated, or coated (where required).	X	
	Area of rust.	X	
	Color not as specified, area of no film, sags or runs, foreign matter imbedded in finish, or finish wet or tacky (painted surfaces).		X

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Porcelain enamel finish (see 6.3)	Bare metal shows through on corners.	X	
	Ground coat visible through cover coat on exterior surfaces.	X	
	Chipped, exterior surface or surface likely to contact food.	X	
	Chipped surfaces not likely to contact food.		X
	Fish scaling or burn off.	X	
	Copperhead accompanied by a pulling away or curling up of the enamel.	X	
	Copperheads (small and infrequent specks).		X
	Crazing, crawling, specking or tearing affecting appearance.	X	
	Enamel lumps, wrinkled, or wavy surface.		X
	Hair lines or star marks.		X
Construction and workmanship (general)	Any component missing.	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim, or grind marks.	X	
	Any component fractured, buckled, warped, loose, kinked, dented, punctured, or malformed.	X	
	Top not in line with front and sides.		X
	Top surface of heating units not flat (check with straight edge).		X
Reflectors or drip trays (as applicable)	Not removable.		X
	Dented or bent.		X

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Oven	Broiler pan or rack not supplied or not type specified (when required).	X	
	Oven light not protected.	X	
	Less than two baking racks.		X
	Rack jams when inserted or withdrawn in shelf support.		X
	Separate light switch not provided when door oven has window.	X	
	Heating units not as specified.	X	
Surface heating units	Not as specified.	X	
	Not hinged or easily removable and easily replaceable.	X	
	Not removable.	X	
Thermostat	Not as specified.	X	
	Not protected from direct heat, moisture or grease.	X	
Oven time control	Not as specified.		X
Doors and handles	Openings around doors not in vertical and horizontal alignment.	X	
	Doors for type II ovens not recessed.	X	
	Sides do not project past oven door.	X	
	Handles mounted crooked.		X
	Door will not remain in full open position.	X	
	Door operation tight.		X
	Oven door does not close tight (cannot be adjusted).	X	
	Oven door not rigid when in the full open position.		X

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification Major	Minor
Wiring and switches	Switches broken or jammed.	X	
	Surface switches mounted on cooking top.	X	
	Not at least 6-position 5-heat and one-off.		X
	Push-on connectors not properly attached to the conductors or wrong type of wire.	X	
	Position not marked as specified.	X	
	Loose wiring connection.	X	
	Means to prevent accidental turn-on not provided.	X	
Marking for identification	Missing, incomplete, illegible, not affixed in the prescribed manner, not fabricated of specified material.		X
Instructions (when applicable)	Missing, incomplete, or illegible.	X	

4.3.3.2 Dimensional examination. Examination shall be made of the ranges for compliance with dimensions and oven and storage compartment volumes specified. Any nonconformance to specified requirements shall constitute a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.3.3 Testing of the end item. When no preproduction sample is required one unit from the initial production lot shall be tested in accordance with 4.4.2. Any nonconformance shall be cause for rejection of the lot. Each production unit shall be tested as specified in 4.4.1 and any nonconformance shall be cause for rejection of the affected item.

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4.3.4 Examination of preparation for delivery. An examination shall be made to determine compliance with requirements for preparation for delivery and shall be performed in accordance with MIL-R-11337.

4.3.5 Standards compliance. Proof of compliance with UL and ANSI requirements of 3.3 shall be submitted to the Government representative.

4.4 Tests.

4.4.1 Operational tests. The end item shall be tested for operation and fit of all controls and moving parts. The unit shall be connected to a rated power source and operated without load. During this test run, the test operator shall make an examination of all equipment and controls properly connected to each position of the surface unit switches, and to each position of the oven and broiler control, as applicable. All manually operated controls shall be operated through a complete cycle to determine compliance with 3.4.11. When applicable, the internal timer shall be set and the alarm signal tested to determine compliance with 3.4.2.4 and 3.5.1.3. Any nonconformance to the operational requirements specified shall constitute failure of this test.

4.4.2 Surface temperatures. Surface temperatures on the door handles, knobs, top, sides and door panel of the ranges shall be tested to determine conformance to the requirements of 3.4.4 and table I. The thermostat shall be set to maintain an oven temperature of 330 F. $\pm$ 5 F., above room temperature. Temperatures shall be taken one hour after a constant oven temperature has been attained. Temperatures shall be determined by use of an indicating surface pyrometer or thermocouples securely fastened to the surface being measured. One reading shall be taken on the grip area of the door handles and knobs. The top, sides, and door panels (excluding glazed area) shall be divided into 8 equal areas with readings taken at the center of the areas. The temperatures taken on each surface shall be averaged to determine compliance. Any nonconformance to the surface temperature requirements specified shall constitute failure of this test.

4.4.3 Plastic drop test. Plastic knob(s) and handle(s) shall be dropped three times from a height of four feet onto an asphalt tile surface over cement floor to determine compliance with 3.4.9.1. This test is to determine acceptability of plastic material and is not intended to cover repairable denting of trim on knobs and handles. Any nonconformance to the requirements specified shall constitute failure of this test.

4.4.4 Porcelain enamel thickness. Porcelain enamel thickness shall be measured with a magnetic thickness gage calibrated to an accuracy of plus or minus 0.001 inch to determine compliance with 3.4.10. Any nonconformance to the requirements specified shall constitute failure of this test.

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5. PREPARATION FOR DELIVERY .

5.1 Levels A, B and C. Preservation, packaging, and packing for each level of protection shall be in accordance with the applicable requirements of MIL-R-11337 (see 6.2).

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-R-11337.

6. NOTES

6.1 Intended use. The ranges are intended for use in domestic kitchens of civilian and Military quarters and other operational type kitchens for which these ranges are best suited, i.e., fire departments, substations, etc.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, style (if applicable) and size of range required (see 1.2.1).
(c) First article. When a preproduction sample is not required (see 3.1).
(d) Electrical characteristics, if other than specified (see 3.4.1).
(e) When a heat resisting viewing window door is required (see 3.4.2.1).
(f) When oven light is not required (see 3.4.2.2).
(g) When oven time-control is not required (see 3.4.2.4).
(h) When manuals for civil agencies are other than specified (see 3.4.5.1).
(i) When quantity of manuals and model number requirements on Military procurements are other than specified (see 3.4.5.2, 3.4.5.2.1, and 3.4.5.2.2).
(j) When an indicating light is not required (see 3.4.6).
(k) When three surface units are required for size 4 or 5 ranges (see 3.5.1.1 and 3.5.2.1).
(l) When a thermostatically controlled surface unit is required for size 2 or 3 or style 2 ranges (see 3.5.1.1 and 3.5.2.1).

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(m) When additional appliance recepticals are required (see 3.5.1.4).
(n) When an appliance outlet is required for size 4 or 5 range (see 3.5.1.4).
(o) When appliance receptical(s) is required for type II ranges (see 3.5.2.4).
(p) Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Definitions. For definitions of porcelain enamel defects consult ASTM Publication C 286 - Terms Relating to Porcelain Enamel.

6.4 Supersession data. Items identified herein as "Styles" and "sizes" were listed under "Type I and Type II" of Federal Specification W-R-101e. Items identified as type III and IV ranges of Federal Specification W-R-101e were deleted from this specification.

MILITARY CUSTODIANS:

Army - GL
Navy - YD
Air Force - 84

Preparing activity:

Army - GL

Project No. 7310-0250

Review activities:

Army - MD
Navy - MC, YD
Air Force - 84

User activity:

Army - CE

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